Supplementary Material

LAMP-Merge: Long-tail-Aware Medical Prototype Merging for One-shot Asynchronous Clinical Models

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A Overview of the Supplementary Material

This supplement is intended to make the empirical and theoretical claims in the paper reproducible without repeating the main exposition. It contains four types of material. First, Section B gives an executable-level specification of LAMP-Merge, emphasizing the boundary between local hospital computation and server-side merging. Second, Sections C and D provide additional experimental context: the full Organ-S results omitted from the compact main tables, and the class-wise dataset statistics used to construct the reported client splits. Third, Section E defines every internal ablation by the exact mathematical substitution applied to the full method, so that each ablation tests a single modeling choice. Finally, Sections F–H provide supporting theory, metric definitions, and hyperparameter-sweep details required for reproduction and checklist verification.

Unless otherwise stated, K denotes the number of clients, C denotes the number of diagnosis classes, $n_{i,c}$ denotes the support count of class c on client i, $m_{i,c}$ denotes the corresponding prevalence count, and $\phi_0(T(\cdot))$ denotes the frozen reference encoder applied after the evaluation transform T.

B Algorithmic Specification

Algorithm 1 provides the procedural specification corresponding to the method section of the paper. It is included here to clarify the execution order and the information boundary, rather than to introduce an additional variant of LAMP-Merge. The inputs are the local datasets $\{D_i\}_{i=1}^K$, the shared frozen reference encoder ϕ_0, the evaluation transform T, and four scalar hyperparameters γ, s, τ, λ. The output is a single merged diagnostic model $\theta_m = (\phi_0, \{w_c, b_c\}_{c=1}^C)$.

The algorithm separates local statistic construction from server-side merging. Lines 1–6 are executed independently by each hospital. Client i computes the class prototype $\mu_{i,c}$, support count $n_{i,c}$, and prevalence count $m_{i,c}$ for every observed diagnosis class. The upload contains only these class-level statistics. Lines 7–20 are executed once on the server. The server first reconstructs a class-specific prototype p_c through support-aware aggregation and then converts it into the classifier weight w_c. The prevalence counts are used only to form the gated centered log-prior b_c. Thus, the merged model is obtained in one pass, without raw-image transfer, gradient exchange, or additional local optimization after the upload.

Algorithm 1 LAMP-Merge

Require: Clients $\{D_i\}_{i=1}^K$; frozen reference encoder ϕ_0; transform T; hyperparameters γ, s, τ, λ
Ensure: Merged diagnostic model $\theta_m = (\phi_0, \{w_c, b_c\}_{c=1}^C)$
1: **Hospital side** (each client i, uploaded once):
2: **for** each class $c = 1, \dots, C$ **do**
3: $\quad n_{i,c} \leftarrow |D_{i,c}|$; $\quad m_{i,c} \leftarrow$ local label count of class c
4: $\quad \mu_{i,c} \leftarrow \frac{1}{n_{i,c}} \sum_{(x,y)\in D_i} \mathbf{1}[y = c]\, \phi_0(T(x))$ **if** $n_{i,c} > 0$
5: **end for**
6: upload $\{(\mu_{i,c}, n_{i,c}, m_{i,c})\}_{c=1}^C$; **raw images stay local**
7: **Merging** (single pass):
8: **for** each class $c = 1, \dots, C$ **do**
9: $\quad e_{i,c} \leftarrow (n_{i,c}+1)^\gamma\, \mathbf{1}[n_{i,c} > 0]$ for all i
10: $\quad \alpha_{i,c} \leftarrow e_{i,c} / \sum_{j=1}^K e_{j,c}$
11: $\quad p_c \leftarrow \sum_{i=1}^K \alpha_{i,c}\, \mu_{i,c}$ $\triangleright$ DPR fusion
12: $\quad w_c \leftarrow s\, p_c / \|p_c\|_2$
13: **end for**
14: $\pi_c \leftarrow \sum_i m_{i,c} / \sum_k \sum_i m_{i,k}$; $\quad r \leftarrow C \max_c \pi_c$
15: **if** $r > \tau$ **then**
16: $\quad b_c \leftarrow \lambda\big(\log \pi_c - \frac{1}{C}\sum_k \log \pi_k\big)$ for all c $\triangleright$ LPC
17: **else**
18: $\quad b_c \leftarrow 0$ for all c
19: **end if**
20: **return** $\theta_m = (\phi_0, \{w_c, b_c\}_{c=1}^C)$

The algorithm also specifies the privacy and communication assumptions used throughout the experiments. The client upload is $\{(\mu_{i,c}, n_{i,c}, m_{i,c})\}_{c=1}^C$; it does not contain raw images, sample identities, gradients, or a server-side validation set. Once these statistics and the hyperparameters are fixed, the server-side computation is deterministic. This makes the merge reproducible from the uploaded statistics alone and distinguishes the setting from multi-round federated optimization.

C Additional Organ-S Results

The main paper reports the compact set of benchmark tables required to compare methods across model families. To keep

Table 1: Organ-S client-average test ACC / Macro-F1 (%) on Swin-Tiny.

Method	$K = 3$	$K = 5$	$K = 7$	Avg
Weight Avg.	23.05 / 12.58	17.70 / 9.52	19.07 / 10.37	19.94 / 10.82
TIES	14.68 / 8.88	14.03 / 9.87	15.91 / 10.90	14.87 / 9.88
DARE-Linear	20.61 / 11.41	16.22 / 9.41	15.52 / 8.33	17.45 / 9.72
DARE-TIES	14.24 / 8.21	12.98 / 8.61	11.15 / 7.05	12.79 / 7.96
RegMean	20.04 / 10.36	20.95 / 9.04	19.66 / 10.78	20.22 / 10.06
Fisher	18.47 / 11.70	17.36 / 8.75	20.49 / 10.35	18.77 / 10.27
Breadcrumbs	17.67 / 10.96	12.37 / 8.14	11.13 / 7.21	13.72 / 8.77
Model Stock	21.86 / 12.85	17.11 / 9.26	17.15 / 8.73	18.71 / 10.28
Iso-C	25.18 / 15.34	22.82 / 10.93	18.34 / 10.30	22.12 / 12.19
Free-Merging	23.05 / 12.58	17.70 / 9.52	19.07 / 10.37	19.94 / 10.82
RobustMerge	17.85 / 10.64	16.22 / 10.64	15.42 / 8.45	16.50 / 9.91
FROM	18.11 / 10.16	17.43 / 9.56	12.89 / 7.62	16.14 / 9.11
LAMP-Merge	**62.58 / 58.21**	**62.80 / 58.45**	**62.84 / 58.39**	**62.74 / 58.35**

Table 2: Organ-S client-average test ACC / Macro-F1 (%) on ConvNeXt.

Method	$K = 3$	$K = 5$	$K = 7$	Avg
Weight Avg.	17.66 / 8.18	12.61 / 7.24	14.33 / 5.48	14.87 / 6.97
TIES	11.83 / 6.50	13.13 / 8.79	11.34 / 7.02	12.10 / 7.44
DARE-Linear	17.48 / 8.29	8.18 / 6.03	8.73 / 5.12	11.46 / 6.48
DARE-TIES	14.34 / 7.74	9.80 / 7.03	12.17 / 6.83	12.11 / 7.20
RegMean	15.71 / 9.76	13.87 / 7.28	16.55 / 6.68	15.37 / 7.91
Fisher	22.16 / 10.64	16.09 / 9.05	17.67 / 8.49	18.64 / 9.39
Breadcrumbs	9.86 / 6.89	7.97 / 6.03	8.10 / 5.06	8.64 / 5.99
Model Stock	15.72 / 7.26	10.61 / 7.02	11.48 / 5.24	12.60 / 6.50
Iso-C	16.66 / 8.25	12.72 / 8.92	14.98 / 6.69	14.79 / 7.95
Free-Merging	17.66 / 8.18	12.61 / 7.24	14.33 / 5.48	14.87 / 6.97
RobustMerge	13.57 / 6.67	6.79 / 5.45	7.52 / 4.47	9.29 / 5.53
FROM	16.19 / 7.85	14.36 / 6.72	11.94 / 5.65	14.16 / 6.74
LAMP-Merge	**60.55 / 56.06**	**60.50 / 56.07**	**60.66 / 56.09**	**60.57 / 56.07**

the main body readable, the Organ-S table is moved to this supplement while preserving the same evaluation protocol, the same client counts, and the same reporting format. Tables 1–4 report client-average ACC / Macro-F1 in percentage points. **Across all four backbones, LAMP-Merge remains the strongest method on Organ-S, confirming that the conclusion does not depend on omitting this CT view from the compact main tables.**

D Dataset Details

Table 3: Organ-S client-average test ACC / Macro-F1 (%) on ResNet.

Method	$K = 3$	$K = 5$	$K = 7$	Avg
Weight Avg.	23.64 / 7.49	12.41 / 3.79	15.30 / 3.55	17.12 / 4.95
TIES	13.07 / 3.70	11.30 / 4.36	13.47 / 4.43	12.62 / 4.16
DARE-Linear	23.58 / 7.15	13.95 / 4.22	16.91 / 5.20	18.15 / 5.52
DARE-TIES	14.00 / 5.08	9.03 / 3.10	12.42 / 4.16	11.81 / 4.11
RegMean	20.27 / 9.88	9.12 / 4.70	11.81 / 5.89	13.73 / 6.82
Fisher	16.69 / 5.25	10.74 / 4.53	14.84 / 3.24	14.09 / 4.34
Breadcrumbs	13.20 / 7.20	12.05 / 4.78	11.83 / 4.58	12.36 / 5.52
Model Stock	23.48 / 7.35	12.63 / 4.26	15.25 / 3.61	17.12 / 5.07
Iso-C	25.31 / 10.64	10.91 / 5.37	13.97 / 4.49	16.73 / 6.83
Free-Merging	23.64 / 7.49	12.41 / 3.79	15.30 / 3.55	17.12 / 4.95
RobustMerge	19.44 / 6.52	10.32 / 4.39	11.86 / 4.03	13.87 / 4.98
FROM	16.73 / 4.99	12.10 / 3.48	12.30 / 3.11	13.71 / 3.86
LAMP-Merge	**52.98 / 42.87**	**52.79 / 42.54**	**52.81 / 42.32**	**52.86 / 42.58**

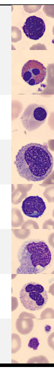
Table 4: Organ-S client-average test ACC / Macro-F1 (%) on ViT-Tiny.

Method	$K = 3$	$K = 5$	$K = 7$	Avg
Weight Avg.	10.06 / 6.70	12.16 / 5.80	9.20 / 3.47	10.47 / 5.32
TIES	8.67 / 4.89	13.92 / 3.58	10.29 / 4.46	10.96 / 4.31
DARE-Linear	8.96 / 5.43	13.59 / 4.62	13.99 / 5.57	12.18 / 5.20
DARE-TIES	8.21 / 5.67	13.62 / 5.14	10.75 / 5.48	10.86 / 5.43
RegMean	13.81 / 8.29	13.39 / 6.79	10.26 / 5.19	12.48 / 6.76
Fisher	12.43 / 6.85	13.30 / 6.28	10.96 / 5.10	12.23 / 6.08
Breadcrumbs	5.26 / 2.62	4.26 / 2.66	4.71 / 3.45	4.74 / 2.91
Model Stock	7.79 / 4.90	8.07 / 4.48	6.81 / 3.01	7.56 / 4.13
Iso-C	10.34 / 7.88	10.48 / 5.48	9.02 / 3.52	9.95 / 5.62
Free-Merging	10.06 / 6.70	12.16 / 5.80	9.20 / 3.47	10.47 / 5.32
RobustMerge	6.91 / 3.22	6.31 / 2.87	5.41 / 2.59	6.21 / 2.89
FROM	10.51 / 6.88	13.70 / 5.25	10.61 / 3.90	11.61 / 5.34
LAMP-Merge	**54.39 / 52.10**	**54.51 / 52.28**	**54.65 / 52.33**	**54.52 / 52.24**

This section provides the dataset-level information needed to reproduce the experimental setting. **BloodMNIST, DermaMNIST, OrganCMNIST, and OrganSMNIST are public MedMNIST v2 datasets (Yang, Shi, and Ni 2023); ChaoShengMNIST is a private eight-class ultrasound subset stored in the same NPZ schema as the public benchmarks.** The five datasets cover microscopy, dermoscopy, CT, and ultrasound while varying the number of classes, visual granularity, and prevalence profile. Tables 5–9 report the exact training, validation, and test split sizes used by the experiments, together with representative training samples for each class.

Blood. BloodMNIST is an eight-class microscopy benchmark for recognizing peripheral blood-cell morphology. Its classes include granulocytes, lymphocytes, platelets, and other hematopoietic cells with visually similar staining pat-

Table 5: Class-wise statistics and representative samples for Blood. The layout expands the corresponding row in the main-paper dataset table: the class-count field is replaced by individual class names. Samples are taken directly from the training split.

Dataset	Class	Train	Val	Test	Total	Sample
Blood	Basophil	852	122	244	1,218	
	Eosinophil	2,181	312	624	3,117	
	Erythroblast	1,085	155	311	1,551	
	Immature granulocytes	2,026	290	579	2,895	
	Lymphocyte	849	122	243	1,214	
	Monocyte	993	143	284	1,420	
	Neutrophil	2,330	333	666	3,329	
	Platelet	1,643	235	470	2,348	

terns. This benchmark emphasizes fine-grained cell morphology; it therefore tests whether a merged classifier preserves class-specific evidence rather than collapsing onto a subset of visually dominant cell types.

Derma. DermaMNIST contains seven dermoscopic skin-lesion categories. It is the most visibly long-tailed benchmark in our suite: melanocytic nevus dominates the sample distribution, while several rare lesion categories have much smaller support. This makes DermaMNIST useful for separating two effects that are easy to conflate under accuracy alone: whether a model truly retains multi-class discrimination, and whether it uses a clinically meaningful prevalence prior.

Organ-C and Organ-S. OrganCMNIST and OrganSMNIST provide 11-way abdominal CT organ recognition in coronal and sagittal views, respectively. They share the same organ label space but differ in anatomical view and spatial appearance. We use them to test whether a method remains stable when the semantic task is fixed but the image geometry and organ visibility change across acquisition planes.

ChaoShengMNIST. ChaoShengMNIST is a private eight-class ultrasound image subset. The source retains anonymized class identifiers, so we report the labels as Class 0–Class 7 rather than assigning unverified clinical names. Compared with microscopy and CT, ultrasound images contain stronger speckle noise, lower contrast, and view-dependent boundaries. This makes ChaoShengMNIST a useful stress test for the prediction-collapse diagnostics and the per-class distribution-gap visualization reported in the main paper.

E Ablation Configurations and Interpretation

This section makes the internal ablation study fully specified. Each ablation changes one mathematical object in the

Table 6: Class-wise statistics and representative samples for Derma, using the same format as Table 5.

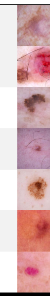
Dataset	Class	Train	Val	Test	Total	Sample
Derma	Actinic keratosis / carcinoma	228	33	66	327	
	Basal cell carcinoma	359	52	103	514	
	Benign keratosis-like lesion	769	110	220	1,099	
	Dermatofibroma	80	12	23	115	
	Melanoma	779	111	223	1,113	
	Melanocytic nevus	4,693	671	1,341	6,705	
	Vascular lesion	99	14	29	142	

Table 7: Class-wise statistics and representative samples for Organ-C.

Dataset	Class	Train	Val	Test	Total	Sample
Organ-C	Bladder	1,148	191	828	2,167	
	Femur–left	619	102	431	1,152	
	Femur–right	595	96	421	1,112	
	Heart	600	202	421	1,223	
	Kidney–left	1,088	132	727	1,947	
	Kidney–right	1,170	157	735	2,062	
	Liver	2,986	429	1,835	5,250	
	Lung–left	1,002	347	549	1,898	
	Lung–right	1,022	352	557	1,931	
	Pancreas	1,173	179	750	2,102	
	Spleen	1,572	205	962	2,739	

LAMP-Merge pipeline while keeping the datasets, backbones, client counts, Dirichlet settings, and all remaining implementation choices fixed. The full method attains 62.21% mean ACC under this protocol and is used as the reference point throughout. The goal is not to introduce alternative methods, but to identify which information source is necessary for the final design.

E.1 Diagnostic Prototype Reconstruction

For class c, DPR aggregates reference-space prototypes as

$$p_c = \sum_{i=1}^{K} \alpha_{i,c}\mu_{i,c}, \qquad \alpha_{i,c} = \frac{e_{i,c}}{\sum_j e_{j,c}}. \tag{1}$$

$$e_{i,c} = (n_{i,c}+1)^{\gamma}\mathbf{1}[n_{i,c}>0]. \tag{2}$$

Here $\mu_{i,c}$ is client i's class-conditional reference-space mean and $n_{i,c}$ is its support for that class. The substitutions below test whether the class semantics in $\mu_{i,c}$ and the class-specific

Table 8: Class-wise statistics and representative samples for Organ-S.

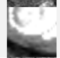
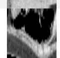
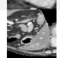
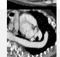
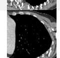
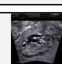
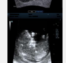
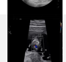
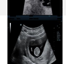
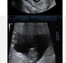
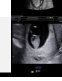
Dataset	Class	Train	Val	Test	Total	Sample
Organ-S	Bladder	1,148	188	811	2,147	
	Femur–left	630	104	439	1,173	
	Femur–right	614	95	445	1,154	
	Heart	721	246	510	1,477	
	Kidney–left	1,132	140	704	1,976	
	Kidney–right	1,119	159	693	1,971	
	Liver	3,464	491	2,078	6,033	
	Lung–left	741	261	397	1,399	
	Lung–right	803	275	439	1,517	
	Pancreas	2,004	280	1,343	3,627	
	Spleen	1,556	213	968	2,737	

Table 9: Class-wise statistics and representative samples for ChaoShengMNIST. The private ultrasound subset retains anonymized class identifiers, which are reported without assigning unverified clinical names.

Dataset	Class	Train	Val	Test	Total	Sample
ChaoSheng	Class 0	420	60	121	601	
	Class 1	419	59	121	599	
	Class 2	408	58	117	583	
	Class 3	630	90	180	900	
	Class 4	345	49	100	494	
	Class 5	489	69	141	699	
	Class 6	671	95	193	959	
	Class 7	487	69	140	696	

reliability in $\alpha_{i,c}$ are both required. For clarity, each ablation is described by **the changed quantity**, **the diagnostic role of the control**, and **the measured effect**.

1) Classifier-head aggregation (CHA). Changed quantity. CHA replaces the reference-space class prototype by the local classifier-head direction:

$$\mu_{i,c} \longrightarrow h_{i,c}. \tag{3}$$

Here $h_{i,c}$ denotes the classifier weight for class c trained on client i, while the aggregation weights remain those of Eq. 1. **Role.** This control evaluates whether independently trained classifier heads already live in a shared semantic coordinate system. **Effect.** CHA obtains $25.63\%/9.34\%$ mean ACC / Macro-F1, $36.58/43.47$ points below LAMP-Merge. The result shows that local classifier-head directions cannot replace reference-space class evidence.

2) Global-feature mean (GFM). Changed quantity. GFM removes class conditioning by assigning the same client feature mean to every class:

$$\mu_{i,c} \longrightarrow g_i, \qquad g_i = \frac{1}{N_i} \sum_{(x,y)\in D_i} \phi_0(T(x)). \tag{4}$$

Here $N_i = \sum_k n_{i,k}$ is the total local sample count. **Role.** This control tests whether a generic medical-domain feature summary is sufficient for merging. **Effect.** GFM obtains $26.50\%/4.47\%$ mean ACC / Macro-F1, $35.71/48.34$ points below LAMP-Merge. The drop indicates that the effective evidence is diagnosis-conditioned, not only domain-conditioned.

3) Support-only synthetic head (SSH). Changed quantity. SSH preserves the cosine-head form and support weights but replaces the aggregated prototype by a data-independent unit vector:

$$p_c \longrightarrow u_c, \qquad \|u_c\|_2 = 1. \tag{5}$$

Role. This control tests whether the gain comes from the classifier parameterization or support weighting alone. **Effect.** SSH obtains $10.89\%/5.77\%$ mean ACC / Macro-F1, $51.32/47.04$ points below LAMP-Merge. Thus, the diagnostic content of the prototype is indispensable.

4) Shuffled-label prototype (SLP). Changed quantity. SLP keeps the prototype set but permutes its class assignment:

$$p_c \longrightarrow p_{\sigma(c)}. \tag{6}$$

Here σ is a fixed permutation over the C diagnosis labels. **Role.** This control tests whether correct prototype-label alignment is necessary beyond geometric separation. **Effect.** SLP obtains $19.57\%/13.10\%$ mean ACC / Macro-F1, $42.64/39.71$ points below LAMP-Merge. The result shows that the reconstructed prototype must be assigned to the correct diagnosis.

5) Uniform client weight (UCW). Changed quantity. UCW keeps $\mu_{i,c}$ unchanged but replaces support-aware reliability with equal weighting among clients that observed the class:

$$e_{i,c} \longrightarrow \mathbf{1}[n_{i,c} > 0]. \tag{7}$$

$$\alpha_{i,c} = \frac{e_{i,c}}{\sum_j e_{j,c}}. \tag{8}$$

Role. This control tests whether class presence alone is sufficient, independent of the amount of class evidence. **Effect.** UCW obtains $58.33\%/48.42\%$ mean ACC / Macro-F1, $3.88/4.39$ points below LAMP-Merge. The result indicates that class-specific support magnitude improves the reliability of the aggregate.

6) Binary support only (BSO). Changed quantity. BSO replaces both DPR support and LPC prevalence counts by class-presence indicators:

$$n_{i,c} \longrightarrow \mathbf{1}[n_{i,c} > 0], \qquad m_{i,c} \longrightarrow \mathbf{1}[n_{i,c} > 0]. \tag{9}$$

Role. This control tests whether fine-grained class counts can be reduced to binary class availability. **Effect.** BSO obtains $55.18\%/49.02\%$ mean ACC / Macro-F1, $7.03/3.79$ points below LAMP-Merge. The result shows that count magnitude matters for both prototype reliability and prevalence calibration.

7) Global client-size weight (GCSW). Changed quantity. GCSW replaces class-specific support with total client size:

$$e_{i,c} \longrightarrow (N_i + 1)^{\gamma}\mathbf{1}[n_{i,c} > 0], \qquad N_i = \sum_k n_{i,k}. \tag{10}$$

Role. This control tests whether a generally large client should be trusted more for every diagnosis. **Effect.** GCSW obtains $59.39\%/48.74\%$ mean ACC / Macro-F1, $2.82/4.07$ points below LAMP-Merge. The result supports class-level reliability: total client size cannot fully replace evidence for the current class.

DPR summary. The semantic controls cause large ACC reductions (35.71–51.32 points), while the weighting controls cause smaller but consistent ACC reductions (2.82–7.03 points). **Together, these results support the two parts of Eq. 1: class-conditioned shared-space evidence provides the discriminative direction, and class-specific support determines how reliably each client contributes to that direction.**

E.2 Long-tail Prevalence Calibration

LPC estimates an empirical prior and adds a centered, gated log-prior bias after DPR:

$$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}}, \tag{11}$$

$$b_c = \lambda \left(\log \pi_c - \frac{1}{C} \sum_k \log \pi_k \right). \tag{12}$$

$$\ell_c(x) = w_c^{\top} \phi_0(T(x)) + \mathbf{1}[r > \tau] b_c. \tag{13}$$

The three controls retain DPR and test the prior estimator or the activation gate in Eq. 11. Each item follows the same format as the DPR ablations.

1) Uniform prevalence prior (UPP). Changed quantity. UPP removes the empirical prevalence signal by replacing the prior with a uniform distribution:

$$\pi_c \longrightarrow \frac{1}{C}. \tag{14}$$

Role. This control tests whether calibration should ignore the observed medical long-tail distribution after DPR has reconstructed the class directions. **Effect.** UPP obtains $58.91\%/53.52\%$ mean ACC / Macro-F1. Its ACC is 3.30 points below LAMP-Merge, while its Macro-F1 is slightly higher. This indicates that a uniform prior can favor rare-class balance, but it sacrifices the prevalence-aware accuracy target of the full model.

2) Always-on calibration (AOC). Changed quantity. AOC removes the long-tail activation gate and applies the prevalence bias to every configuration:

$$\mathbf{1}[r > \tau] \longrightarrow 1. \tag{15}$$

Role. This control tests whether prevalence calibration should be unconditional or activated only when the class distribution is sufficiently skewed. **Effect.** AOC obtains $61.90\%/51.96\%$ mean ACC / Macro-F1, $0.31/0.85$ points below LAMP-Merge. The small but consistent reduction indicates that the gate is a targeted stabilizer rather than the primary source of the performance gain.

3) Client-balanced prior (CBP). Changed quantity. CBP replaces the sample-count empirical prior by equal client influence:

$$\pi_c \longrightarrow \frac{1}{K} \sum_i \frac{m_{i,c}}{\sum_k m_{i,k}}. \tag{16}$$

Role. This control tests whether each client should contribute equally to the prior regardless of its sample volume. **Effect.** CBP obtains $58.99\%/52.40\%$ mean ACC / Macro-F1, $3.22/0.41$ points below LAMP-Merge. The result favors the sample-count prevalence estimate in Eq. 11, because it better reflects the global long-tail distribution.

LPC summary. Every LPC control is below the 62.21% LAMP-Merge mean ACC. UPP and CBP show that both prevalence information and its sample-size weighting matter for the accuracy objective; AOC shows that the activation criterion is a targeted safeguard rather than the primary source of the gain. **Hence LPC calibrates DPR's restored class directions instead of replacing them with a prior.**

F Theoretical Analysis

This section gives the theoretical support for the two design choices used in LAMP-Merge. The first part analyzes Diagnostic Prototype Reconstruction and shows how support-aware aggregation controls the estimation error of the reconstructed class direction. The second part analyzes Long-tail Prevalence Calibration and shows that the centered log-prior acts as a bounded adjustment to the reconstructed head. Throughout the analysis, all features are represented in the shared reference space induced by the frozen encoder $\phi_0(T(\cdot))$, so client prototypes are compared in a common coordinate system.

F.1 Assumptions

Assumption 1 (Within-client sampling). Fix a class c. For every client i with support $n_{i,c} > 0$, the reference features $\{\phi_0(T(x)) : (x,y) \in D_i,\, y = c\}$ are i.i.d. draws from a client-conditional distribution with mean $\mu^{\star}_{i,c} := \mathrm{E}[\phi_0(T(x)) \mid y = c, \text{client } i]$ and covariance $\Sigma_{i,c}$ satisfying $\mathrm{tr}(\Sigma_{i,c}) \le \sigma_c^2$. Features drawn by different clients are mutually independent.

Assumption 2 (Prevalence positivity). The global prior of Eq. 11 satisfies $\pi_c \in [\pi_{\min}, \pi_{\max}] \subseteq (0,1]$ for all c, with $\pi_{\min} > 0$.

The within-client sampling assumption is the standard i.i.d. model applied inside each client-class stratum; cross-client independence reflects that hospitals sample patients separately. It does not assume that client means are identical. The quantity $\mu^{\star}_{i,c}$ may differ across clients because of scanner or population shift, and this discrepancy appears as the bias term below. The prevalence positivity assumption holds whenever every class is observed at the population level; in practice, the additive count in $e_{i,c}$ and the empirical prior keep $\pi_{\min} > 0$.

F.2 Prototype estimation and class margins

Recall from Eq. 1 that the reconstructed prototype is the client mixture $p_c = \sum_i \alpha_{i,c}\mu_{i,c}$, where $\mu_{i,c} = n_{i,c}^{-1}\sum_{y=c}\phi_0(T(x))$ is the empirical client-class mean and $\alpha_{i,c} \geq 0$, $\sum_i \alpha_{i,c} = 1$, with $\alpha_{i,c} = 0$ whenever $n_{i,c} = 0$. Let $\mu^{\star}_c$ denote the target population mean for class c in the reference space, and define the aggregation bias

$$\mathrm{Bias}_c := \Big\| \sum_i \alpha_{i,c}\, \mu^{\star}_{i,c} - \mu^{\star}_c \Big\|_2. \tag{17}$$

Proposition 1 (Bias–variance bound). Under the within-client sampling assumption,

$$\mathrm{E}\big\|p_c - \mu^{\star}_c\big\|_2^2 \;\leq\; \sigma_c^2 \sum_{i:\, n_{i,c}>0} \frac{\alpha_{i,c}^2}{n_{i,c}} \;+\; \mathrm{Bias}_c^2. \tag{18}$$

Proof. Write $p_c - \mu^{\star}_c = (p_c - \mathrm{E}\,p_c) + (\mathrm{E}\,p_c - \mu^{\star}_c)$. Since each $\mu_{i,c}$ is an average of $n_{i,c}$ i.i.d. features, $\mathrm{E}\,\mu_{i,c} = \mu^{\star}_{i,c}$ and $\mathrm{E}\,p_c = \sum_i \alpha_{i,c}\mu^{\star}_{i,c}$, so the second term has norm Bias_c. The two terms are orthogonal in expectation because $\mathrm{E}[p_c - \mathrm{E}\,p_c] = 0$, giving $\mathrm{E}\|p_c - \mu^{\star}_c\|_2^2 = \mathrm{E}\|p_c - \mathrm{E}\,p_c\|_2^2 + \mathrm{Bias}_c^2$. By cross-client independence, the variance term factorizes over clients,

$$\mathrm{E}\|p_c - \mathrm{E}\,p_c\|_2^2 = \sum_i \alpha_{i,c}^2\, \mathrm{E}\|\mu_{i,c} - \mu^{\star}_{i,c}\|_2^2, \tag{19}$$

and for an average of $n_{i,c}$ i.i.d. vectors $\mathrm{E}\|\mu_{i,c} - \mu^{\star}_{i,c}\|_2^2 = \mathrm{tr}(\Sigma_{i,c})/n_{i,c} \leq \sigma_c^2/n_{i,c}$. Terms with $n_{i,c} = 0$ carry $\alpha_{i,c} = 0$ and drop out, yielding (18).

This bound makes the role of the support-aware weight explicit. The variance term $\sigma_c^2 \sum_i \alpha_{i,c}^2/n_{i,c}$ is reduced by assigning larger weights to clients with larger class-specific support. This is precisely the role of $e_{i,c} = (n_{i,c} + 1)^{\gamma}\mathbf{1}[n_{i,c} > 0]$. The indicator also forces $\alpha_{i,c} = 0$ for clients that never observed class c, keeping absent-class noise out of the reconstruction. This is the quantity perturbed by the weighting controls: **UCW** ignores the unequal variances $\sigma_c^2/n_{i,c}$, while **GCSW** weights by total client size $N_i = \sum_k n_{i,k}$ and couples the class-c estimate to data collected for other classes. Their measured drops of 3.88 and 2.82 ACC points are therefore consistent with the variance term in Eq. (18).

We now translate a small prototype error into preserved decision ordering. For a reference feature $z = \phi_0(T(x))$ define the population and reconstructed prototype margins between classes c and d,

$$\Delta_{c,d}(x) = (\mu^{\star}_c - \mu^{\star}_d)^{\top} z, \qquad \widehat{\Delta}_{c,d}(x) = (p_c - p_d)^{\top} z. \tag{20}$$

Proposition 2 (Margin preservation). Fix x with $z \neq 0$ and suppose $\Delta_{c,d}(x) > 0$. If

$$\|p_c - \mu^{\star}_c\|_2 + \|p_d - \mu^{\star}_d\|_2 \;<\; \frac{\Delta_{c,d}(x)}{\|z\|_2}, \tag{21}$$

then $\widehat{\Delta}_{c,d}(x) > 0$; that is, the prototype classifier ranks c above d exactly as the population prototypes do.

Proof. By linearity and the Cauchy–Schwarz inequality,

$$|\widehat{\Delta}_{c,d}(x) - \Delta_{c,d}(x)| = \big|(p_c - \mu^{\star}_c)^{\top} z - (p_d - \mu^{\star}_d)^{\top} z\big|. \tag{22}$$

$$|\widehat{\Delta}_{c,d}(x) - \Delta_{c,d}(x)| \leq \big(\|p_c - \mu^{\star}_c\|_2 + \|p_d - \mu^{\star}_d\|_2\big)\|z\|_2. \tag{23}$$

Under (21) the right-hand side is strictly below $\Delta_{c,d}(x)$, hence $\widehat{\Delta}_{c,d}(x) > \Delta_{c,d}(x) - \Delta_{c,d}(x) = 0$.

The cosine head $w_c = s\, p_c/\|p_c\|_2$ used by LAMP-Merge scores in the direction of p_c, so the margin condition above is the relevant guarantee: when DPR drives the per-class error below the population margin in Eq. (21), the reconstructed head preserves the class ordering. The bias–variance bound explains how class-specific support weighting can reduce this error, and the margin condition explains why the error reduction matters for classification. The semantic controls violate the premise of the argument: **CHA** and **GFM** replace $\mu_{i,c}$ by quantities whose expectation is not $\mu^{\star}_c$, **SSH** discards the class prototype entirely, and **SLP** assigns the prototype to the wrong label. Their large losses of 35.71–51.32 ACC points are therefore aligned with the theory.

F.3 Bounded prevalence correction

We finally show that LPC is a bounded, gauge-fixed adjustment. Recall the centered log-prior bias $b_c = \lambda\big(\log \pi_c - \frac{1}{C}\sum_k \log \pi_k\big)$ from Eq. 11.

Proposition 3 (Zero-sum, invariance, and boundedness). Under the prevalence positivity assumption, the bias satisfies:

(i) *Zero-sum:* $\sum_{c=1}^{C} b_c = 0$, so the correction adds no uniform shift to the logits.

(ii) *Gauge invariance:* $b_c - b_d = \lambda \log(\pi_c/\pi_d)$, independent of the centering constant; adding any constant to all class logits leaves every pairwise comparison, and hence the $\arg\max$, unchanged.

(iii) *Boundedness:* $|b_c| \leq \lambda \log(\pi_{\max}/\pi_{\min})$ and $|b_c - b_d| \leq \lambda \log(\pi_{\max}/\pi_{\min})$ for all c, d.

Proof. (i) $\sum_c b_c = \lambda\big(\sum_c \log \pi_c - C \cdot \frac{1}{C}\sum_k \log \pi_k\big) = 0$. (ii) The centering term is common to b_c and b_d and cancels, leaving $b_c - b_d = \lambda(\log \pi_c - \log \pi_d)$; since softmax and $\arg\max$ depend only on logit differences, the shared constant is irrelevant. (iii) Writing $\bar{\ell} = \frac{1}{C}\sum_k \log \pi_k$, we have $\log \pi_{\min} \leq \bar{\ell} \leq \log \pi_{\max}$, so $|\log \pi_c - \bar{\ell}| \leq \log \pi_{\max} - \log \pi_{\min} = \log(\pi_{\max}/\pi_{\min})$; multiply by λ. The pairwise bound follows from $|\log \pi_c - \log \pi_d| \leq \log(\pi_{\max}/\pi_{\min})$.

Combining the margin condition with the boundedness result shows that the two modules compose without conflict. The score used at test time is $\ell_c(x) = w_c^{\top} z + \mathbf{1}[r >$

$\tau]\, b_c$, so calibration shifts the pairwise margin by at most $|b_c - b_d| \le \lambda \log(\pi_{\max}/\pi_{\min})$. Hence whenever the DPR margin already exceeds this bound, the class ordering is unchanged by LPC:

$$\widehat{\Delta}_{c,d}(x) \;>\; \lambda \log \frac{\pi_{\max}}{\pi_{\min}} \implies \ell_c(x) > \ell_d(x). \tag{24}$$

The correction therefore acts only on near-ties, tipping them toward the more prevalent class, and cannot override a confident reconstructed direction; the gate $\mathbf{1}[r > \tau]$ with $r = C \max_c \pi_c$ additionally switches it off when the configuration is close to balanced. This is the formal counterpart of the empirical finding that a prevalence-only mechanism cannot recover the semantic controls: by (i)–(ii) the bias contributes no class-discriminative direction of its own, and by (iii) its influence is bounded, so it can calibrate the directions DPR supplies but cannot substitute for them.

G Metric Definitions

The main tables report ACC and Macro-F1 as complementary classification metrics. ACC is the fraction of correctly classified test samples,

$$\mathrm{ACC} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}[\hat{y}_n = y_n], \tag{25}$$

while Macro-F1 averages the per-class F1 scores and therefore gives rare classes the same weight as majority classes. AUC is used in the radar summaries when probabilistic outputs are available; it measures ranking quality from class probabilities and is less tied to a single operating threshold than ACC.

The prediction-collapse metrics are defined on the predicted class distribution:

$$q_c = \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}[\hat{y}_n = c]. \tag{26}$$

The collapse ratio measures the largest predicted-class share:

$$\rho = \max_c q_c. \tag{27}$$

The effective number of predicted classes measures how many classes are actively used by the model:

$$C_{\text{eff}} = \exp\left(-\sum_c q_c \log q_c\right). \tag{28}$$

The total variation distance compares the predicted distribution q with the empirical test distribution $\bar{q}$:

$$\mathrm{TV}(q, \bar{q}) = \frac{1}{2} \sum_c |q_c - \bar{q}_c|. \tag{29}$$

Lower ρ and TV and higher C_{eff} indicate weaker collapse. The absolute per-class gaps visualized in the main paper are $|q_c - \bar{q}_c|$ in percentage points and expose which classes contribute to TV.

The long-tail metrics are defined on empirical class counts rather than model predictions. For class counts N_c, the empirical prevalence is

$$\bar{p}_c = \frac{N_c}{\sum_k N_k}. \tag{30}$$

The majority prevalence and imbalance ratio are

$$\bar{p}_{\max} = \max_c \bar{p}_c, \qquad R_{\text{imb}} = \frac{\max_c N_c}{\min_c N_c}. \tag{31}$$

These data-side quantities quantify whether the dataset itself is skewed. They are used only to justify prevalence-aware calibration; they are not treated as evidence that a merged model is correct. This distinction is important: a high ρ is a model failure when it arises from collapse, whereas a high $\bar{p}_{\max}$ is a property of the dataset that LPC must preserve in a bounded manner.

H Hyperparameter-Sweep Reproducibility

Every point in the hyperparameter-sensitivity study of the main paper is a completed **full-scope experiment** covering five datasets, four backbones, three client counts, and three Dirichlet settings. The plotted curves retain measured points in ascending parameter order; no smoothing, interpolation, or best-run selection is applied. The LPC grid records $\tau \in \{2.0, 2.5, 3.0\}$ and calibration strengths $\lambda \in \{3.00, 3.25, \ldots, 5.25\}$, with $\gamma = 0.55$ and $s = 18.75$ fixed. DPR completion records are plotted only after the full grid point has succeeded. The selected operating point is $\gamma = 0.55$, $s = 18.75$, $\tau = 2.5$, and $\lambda = 4.25$, making its reported value traceable to the same completed grid as its neighboring points.

References

Yang, J.; Shi, R.; and Ni, B. 2023. MedMNIST v2: A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification. *Scientific Data*, 10(1): 41.